

Bent Marshak Waves - Hurricane & Hammer

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1 Introduction

- Radiation-driven heat waves (Marshak waves) appear in HEDP, ICF, and astrophysical systems.
- Classical Marshak theory describes 1D planar diffusion with $\sqrt{t}$ scaling.
- Experiments show curved radiation fronts in foam targets.
- Curvature is caused primarily by energy loss into surrounding walls.
- **Goal: derive an analytic 2D model for bent Marshak waves with lossy boundaries.**

2 Boundaries Problem

Physical setup

- 2D planar slab geometry - working in Cartesian coordinates (x, y) .
- Constant temperature source T_s at $x = 0$.
- Radiation front at $x = x_F(y, t)$.
- Lossy walls at $y = \pm L$ with albedo a - constant over time in the article.
- The article derive the model assuming LTE, meaning $T_{rad} = T_{mat} = T$ (black).
- Another strong assumption is that the energy (e) and opacity (κ) are constant in T .
- The fundamental equation is:

$$\rho \frac{\partial e}{\partial t} = \frac{4}{3} \nabla \cdot \left(\frac{1}{\kappa \rho} \nabla (\sigma_{SB} T^4) \right) \quad (1)$$

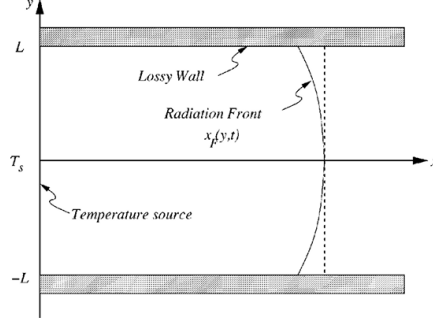


Figure 1: The physical setup of a bent Marshak wave propagating in a 2D slab with lossy walls - Cartesian coordinates (x, y) .

Governing equation

- Supersonic assumption: material motion negligible, meaning the material stays in place over times scales of interest, so ρ is constant in time and space.
- Quasi-steady assumption: time derivative small compared to spatial diffusion, so $\partial_t e$ negligible.
- Radiation diffusion equation reduces to:

$$\nabla^2 T^4 = 0$$

- Temperature field expanded in eigenmodes - Laplace equation in Cartesian coordinates.
- General solution: define $u = T^4$ then we get $\nabla^2 u = u_{xx} + u_{yy} = 0$. Using separation of variables $u(x, y) = X(x)Y(y)$ we get:

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = k^2$$

So we get:

$$X(x) = Ae^{kx} + Be^{-kx}$$

$$Y(y) = C \cos(ky) + D \sin(ky)$$

Therefore the general solution is (knowing the problem symmetry around $y = 0$):

$$T^4(x, y) = \sum_{n=0}^{\infty} [A_n e^{k_n x} + B_n e^{-k_n x}] \cos(k_n y) \quad (2)$$

Notice that the heat-wave front depends only on y and t , and knowing the y -symmetry we can write it as:

$$x_F(y, t) = \sum_{n=0}^{\infty} c_n(t) \cos(k_n y) \quad (3)$$

where x_F defined by the place where $T(x_F, y, t) = 0$, and so he shares the same eigenvalues k_n as the temperature field.

Wall boundary condition

- Nonideal wall introduces energy loss. Radiation flux F_{in} arrives, but only a fraction a is reflected back as $F_{out} = aF_{in}$. Therefore the flux at the wall is:

$$F_{wall} = F_{in} - F_{out} = (1 - a)F_{in}$$

Now, knowing that $F_{in} = \sigma_{SB}T^4$, we get:

$$F_{wall} = (1 - a)\sigma_{SB}T^4$$

But also, from fick's law we know that:

$$F_{wall} = \mathbf{F} \cdot \hat{y} = -\frac{4}{3\kappa\rho} \frac{\partial(\sigma_{SB}T^4)}{\partial y}$$

Equating the two expressions for F_{wall} we get the boundary condition at $y = \pm L$:

$$\frac{\partial T^4}{\partial y}|_{y=\pm L} = \frac{3}{4}\kappa\rho(a - 1)T^4|_{y=\pm L} \quad (4)$$

Notice that since $a < 1$ we have $(a - 1) < 0$ meaning that the derivative is negative, so the temperature decreases towards the wall, as expected.

Eigenmode expansion

- Radiation temperature expanded as:

$$T^4(x, y) = \sum_n \cos(k_n y) [A_n(t)e^{k_n x} + B_n(t)e^{-k_n x}]$$

Spatial derivative:

$$\frac{\partial T^4}{\partial y} = -\sum_n k_n \sin(k_n y) [A_n e^{k_n x} + B_n e^{-k_n x}]$$

Substitution of mode expansion gives:

$$-k_n \sin(k_n L) = \frac{3}{4}\rho\kappa(a - 1) \cos(k_n L)$$

Rearranged to:

$$\tan(k_n L) = \frac{\varepsilon}{k_n L} \quad (5)$$

Definition of small parameter:

$$\varepsilon = \frac{3}{4}(1-a)\tau$$

where $\tau = \rho\kappa L$ is the optical depth.

- Solutions are intersections of: $y = \tan(k_n L)$ and $y = \frac{\varepsilon}{k_n L}$. For small ε :

$$k_n L \approx n\pi$$

with small corrections.

Physical meaning of ε

- Interpretation:

$$\varepsilon \sim \frac{\text{energy lost sideways per diffusion step}}{\text{energy transported forward}}$$

- Regimes:

- $\varepsilon \ll 1$: weak wall losses $\Rightarrow$ nearly 1D Marshak wave
- $\varepsilon \sim 1$: strong lateral losses $\Rightarrow$ strong bending

Constraint on albedo

- Small-parameter condition:

$$\varepsilon < 1 \quad \Rightarrow \quad a > 1 - \frac{4}{3\tau}$$

- For typical $\tau \sim 3$:

$$a > 1 - \frac{4}{9} \approx 0.56 \quad \Rightarrow \quad \frac{2}{3} \lesssim a < 1$$

- Meaning: theory valid only for **high-albedo walls**.

Eigenvalue expansion

From the wall boundary condition (Eq. 5), we write the eigenvalues as perturbations around the ideal values:

$$\tan(k_n L) = \frac{\varepsilon}{k_n L}, \quad (6)$$

with small correction $\phi_n \ll 1$.

Using the Taylor expansion,

$$\tan(n\pi + \phi_n) \simeq \phi_n + \frac{1}{3}\phi_n^3 + \mathcal{O}(\phi_n^5), \quad (7)$$

the eigenvalue equation becomes

$$\phi_n + \frac{1}{3}\phi_n^3 \simeq \frac{\varepsilon}{n\pi + \phi_n}.$$

To leading order, this gives the quadratic approximation

$$\phi_n^2 + n\pi\phi_n - \varepsilon \simeq 0,$$

with solution

$$\phi_n \simeq \frac{-n\pi + \sqrt{n^2\pi^2 + 4\varepsilon}}{2}. \quad (8)$$

Expanding the square root,

$$\sqrt{n^2\pi^2 + 4\varepsilon} \simeq n\pi \left(1 + \frac{2\varepsilon}{n^2\pi^2} - \frac{2\varepsilon^2}{n^4\pi^4} + \dots \right),$$

we obtain

$$\phi_n \simeq \frac{\varepsilon}{n\pi} - \frac{\varepsilon^2}{n^3\pi^3} + \mathcal{O}(\varepsilon^3), \quad (9)$$

and for the lowest mode:

$$\phi_0 = \sqrt{\varepsilon}. \quad (10)$$

The mode amplitudes are expanded in powers of ε :

$$A_n = A_n^{(0)} + \sqrt{\varepsilon} A_n^{(1/2)} + \varepsilon A_n^{(1)} + \dots, \quad (11)$$

$$B_n = B_n^{(0)} + \sqrt{\varepsilon} B_n^{(1/2)} + \varepsilon B_n^{(1)} + \dots. \quad (12)$$

The goal is to solve for the coefficients $A_n^{(i)}$, $B_n^{(i)}$ order-by-order, using the boundary conditions at the source and front.

Source boundary condition ($x = 0$, $T = T_s$)

Let's assume **constant source temperature** T_s at $x = 0$:

$$T_s^4 = T^4(0, y) = \sum_n [A_n + B_n] \cos(k_n y).$$

Using

$$\cos(k_n y) = \cos\left(\frac{n\pi y}{L} + \frac{\phi_n y}{L}\right),$$

and expanding:

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta,$$

we get:

$$\cos(k_n y) = \cos\left(\frac{n\pi y}{L}\right) \cos\left(\frac{\phi_n y}{L}\right) - \sin\left(\frac{n\pi y}{L}\right) \sin\left(\frac{\phi_n y}{L}\right).$$

with small-angle expansions for ϕ_n , we can use tylor expansions:

$$\cos\left(\frac{\phi_n y}{L}\right) \approx 1 - \frac{(\phi_n y)^2}{2L^2} + \frac{(\phi_n y)^4}{24L^4} + \mathcal{O}(\phi_n^6)$$

$$\sin\left(\frac{\phi_n y}{L}\right) \approx \frac{\phi_n y}{L} - \frac{(\phi_n y)^3}{6L^3} + \mathcal{O}(\phi_n^5)$$

Substituting back, we get:

$$\cos(k_n y) \approx \cos\left(\frac{n\pi y}{L}\right) \left[1 - \frac{(\phi_n y)^2}{2L^2} + \frac{(\phi_n y)^4}{24L^4}\right] - \sin\left(\frac{n\pi y}{L}\right) \left[\frac{\phi_n y}{L} - \frac{(\phi_n y)^3}{6L^3}\right].$$

Substituting the expansions for A_n , B_n , and $\cos(k_n y)$ into the source boundary condition, we collect terms order-by-order in ε . At zeroth order:

$$\sum_n \left[A_n^{(0)} + B_n^{(0)}\right] \cos\left(\frac{n\pi y}{L}\right) = T_s^4.$$

Using orthogonality of cosine functions, we find:

$$A_n^{(0)} + B_n^{(0)} = \begin{cases} T_s^4, & n = 0, \\ 0, & n > 0, \end{cases} \quad (13)$$

At order $\sqrt{\varepsilon}$, we have (since $\phi_0 \sim \sqrt{\varepsilon}$ and $\phi_{n>0} \sim \varepsilon$):

$$\sum_n \left[A_n^{(1/2)} + B_n^{(1/2)}\right] \cos\left(\frac{n\pi y}{L}\right) = 0,$$

leading to:

$$A_n^{(1/2)} + B_n^{(1/2)} = 0. \quad (14)$$

At order ε :

$$0 = \sum_n \left(\varepsilon \left[A_n^{(1)} + B_n^{(1)} \right] \cos\left(\frac{n\pi y}{L}\right) - \left[A_n^{(0)} + B_n^{(0)} \right] \frac{(\phi_n y)}{L} \sin\left(\frac{n\pi y}{L}\right) \right) - \varepsilon \frac{y^2}{2L^2} \left[A_0^{(0)} + B_0^{(0)} \right]. \quad (15)$$

But since $\sin(0) = 0$ and $A_{n>0}^{(0)} + B_{n>0}^{(0)} = 0$, the second term vanishes, leaving:

$$\sum_n \left[A_n^{(1)} + B_n^{(1)} \right] \cos\left(\frac{n\pi y}{L}\right) = \frac{y^2}{2L^2} T_s^4.$$

Now using orthogonality again, we find:

$$A_n^{(1)} + B_n^{(1)} = \begin{cases} \frac{T_s^4}{3}, & n = 0, \\ \frac{2(-1)^n T_s^4}{n^2 \pi^2}, & n > 0. \end{cases} \quad (16)$$

Notice that,

$$\int_{-L}^L y^2 \cos\left(\frac{n\pi y}{L}\right) dy = \begin{cases} \frac{2L^3}{3}, & n = 0, \\ \frac{4(-1)^n L^3}{n^2 \pi^2}, & n > 0. \end{cases}$$

Thus,

$$A_n + B_n = \begin{cases} T_s^4(1 + \frac{\varepsilon}{3}) + \mathcal{O}(\varepsilon^{3/2}), & n = 0, \\ \frac{4(-1)^n \varepsilon T_s^4}{n^2 \pi^2} + \mathcal{O}(\varepsilon^{3/2}), & n > 0. \end{cases}$$

Front boundary condition ($T(x_F, y) \approx 0$)

The problem: the front position $x_F(y, t)$ is unknown a priori, and the temperature must vanish there:

$$T^4(x_F, y) = 0.$$

We use a **weak-curvature approximation**, assuming the front is nearly planar with small transverse perturbations:

$$\delta x(y, t) \sim \mathcal{O}(\varepsilon).$$

Using Eq. 3, we write:

$$x_F(y, t) = c_0(t) + \delta x(y, t), \quad (17)$$

with

$$\delta x(y, t) = \sum_{n \geq 1} c_n(t) \cos(k_n y). \quad (18)$$

Taylor expanding the temperature around $x = c_0(t)$:

$$T^4(x_F, y) \approx T^4(c_0, y) + (x_F - c_0) \partial_x T^4|_{c_0}. \quad (19)$$

But, the LHS must vanish for all y , so to leading order we impose:

$$T^4(c_0, y) \approx [c_0 - \sum_{n=0}^{\infty} c_n \cos(k_n y)] \partial_x T^4|_{c_0} = 0.$$

Or equivalently,

$$T^4(c_0, y) = - \sum_{n=1}^{\infty} c_n \cos(k_n y) \partial_x T^4|_{c_0}.$$

Using the mode expansion

$$T^4(x, y) = \sum_n \cos(k_n y) (A_n e^{k_n x} + B_n e^{-k_n x}),$$

we define

$$D_n(x) \equiv A_n e^{k_n x} + B_n e^{-k_n x}.$$

Thus,

$$T^4(c_0, y) = \sum_n D_n(c_0) \cos(k_n y),$$

$$\partial_x T^4(c_0, y) = \sum_n k_n \cos(k_n y) (A_n e^{k_n c_0} - B_n e^{-k_n c_0}).$$

order by order front condition

The physical front condition is

$$T(x_F, y) \approx 0,$$

which implies, at leading order,

$$T^4(c_0, y) = \mathcal{O}(\varepsilon).$$

Therefore,

$$D_n^{(0)}(c_0) = A_n^{(0)} e^{k_n c_0} + B_n^{(0)} e^{-k_n c_0} = 0. \quad (20)$$

And similarly for $\mathcal{O}(\sqrt{\varepsilon})$:

$$D_n^{(1/2)}(c_0) = A_n^{(1/2)} e^{k_n c_0} + B_n^{(1/2)} e^{-k_n c_0} = 0. \quad (21)$$

This gives the fundamental relation:

$$A_n^{(0)} = -B_n^{(0)} e^{-2k_n c_0}.$$

And similarly for $\mathcal{O}(\sqrt{\varepsilon})$:

$$A_n^{(1/2)} = -B_n^{(1/2)} e^{-2k_n c_0}.$$

At the next order, $\mathcal{O}(\varepsilon)$, we have (same derivation as we did for the source BC with the orders):

$$\begin{aligned} & \varepsilon \sum_{n=0}^{\infty} \left(\varepsilon \left[A_n^{(1)} e^{k_n c_0} + B_n^{(1)} e^{-k_n c_0} \right] \cos\left(\frac{n\pi y}{L}\right) \right. \\ & \quad \left. - \left[A_n^{(0)} e^{k_n c_0} + B_n^{(0)} e^{-k_n c_0} \right] \frac{(\varepsilon y^2)}{2L^2} \right) \\ & = -c_1 \cos\left(\frac{\pi y}{L}\right) \frac{\sqrt{\varepsilon}}{L} \left[A_0^{(0)} e^{k_0 c_0} + B_0^{(0)} e^{-k_0 c_0} \right] \cos\left(\frac{\sqrt{\varepsilon}}{L} y\right). \end{aligned} \quad (22)$$

The second term vanishes due to the leading-order front condition, using orthogonality we get:

$$A_m^{(1)} e^{k_m c_0} + B_m^{(1)} e^{-k_m c_0} = \begin{cases} -c_1 \frac{\sqrt{\varepsilon}}{L\pi^2} \left(A_0^{(0)} e^{k_0 c_0} - B_0^{(0)} e^{-k_0 c_0} \right), & n = 0, \\ -c_1 \frac{(-1)^m \sqrt{\varepsilon} \pi (1+m^2)}{L(m-1)^2(m+1)^2} \left(A_0^{(0)} e^{k_0 c_0} - B_0^{(0)} e^{-k_0 c_0} \right), & n > 0. \end{cases} \quad (23)$$

And this is right to order $\mathcal{O}(\varepsilon)$. Since the RHS is $\mathcal{O}(\sqrt{\varepsilon})$, we can use the leading-order front condition to write:

$$A_n e^{k_n c_0} + B_n e^{-k_n c_0} = 0 + \mathcal{O}((\varepsilon)^{3/2}), \quad (24)$$

which implies that the curvature corrections enter only at order $\mathcal{O}((\varepsilon)^{3/2})$. Furthermore, this yields the fundamental relation:

$$A_n = -B_n e^{-2k_n c_0} + \mathcal{O}((\varepsilon)^{3/2}).$$

Define

$$S_n \equiv A_n + B_n.$$

One can then solve for A_n, B_n :

$$B_n = \frac{S_n e^{k_n c_0}}{2 \sinh(k_n c_0)}, \quad A_n = -\frac{S_n e^{-k_n c_0}}{2 \sinh(k_n c_0)}.$$

Hence,

$$A_n e^{k_n x} + B_n e^{-k_n x} = -S_n \frac{\sinh[k_n(x - c_0)]}{\sinh(k_n c_0)}.$$

Substituting the previously derived coefficients:

$$S_n = \begin{cases} T_s^4 \left(1 + \frac{\varepsilon}{3} \right) + \mathcal{O}(\varepsilon^{3/2}), & n = 0, \\ \frac{4(-1)^n \varepsilon T_s^4}{n^2 \pi^2} + \mathcal{O}(\varepsilon^{3/2}), & n > 0, \end{cases}$$

gives:

$$A_n e^{k_n x} + B_n e^{-k_n x} = \begin{cases} -T_s^4 \left(1 + \frac{\varepsilon}{3} \right) \frac{\sinh(k_0(x - c_0))}{\sinh(k_0 c_0)}, & n = 0, \\ -c_1 \frac{4(-1)^n \varepsilon T_s^4}{n^2 \pi^2} \frac{\sinh(k_0(x - c_0))}{\sinh(k_0 c_0)}, & n > 0. \end{cases} \quad (25)$$

And finally, the reconstructed temperature field is

$$\begin{aligned} \frac{T^4(x, y)}{T_s^4} &= -\left(1 + \frac{\varepsilon}{3} \right) \cos\left(\frac{\sqrt{\varepsilon}}{L} y \right) \frac{\sinh[k_0(x - c_0)]}{\sinh(k_0 c_0)} \\ &+ \frac{4\varepsilon}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos(k_n y) \frac{\sinh[k_n(x - c_0)]}{\sinh(k_n c_0)} + \mathcal{O}(\varepsilon^{3/2}). \end{aligned} \quad (26)$$

At first glance, it may seem atrange to see *varepsilon* both in the eigenvalues k_n and as a prefactor in the expansion. However, this gives the write boundary condition at the source:

$$\left. \frac{T^4}{T_s} \right|_{y=0} = 1 + \frac{\varepsilon}{3} + \frac{4\varepsilon}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = 1,$$

where we used the fact that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = -\frac{\zeta(2)}{2} = -\frac{\pi^2}{12}$.

3 Front Equation of Motion

The dynamics of the radiation front are obtained from local energy balance at the front. The physical statement is:

$$\text{Energy density} \times \text{front velocity} = \text{radiative energy flux into the front}.$$

Mathematically,

$$\rho e \dot{x}_F(y, t) = F_{\text{rad}}(x_F), \quad (27)$$

where ρe is the volumetric internal energy of the cold material, and the radiative flux is given by diffusion theory:

$$F_{\text{rad}} = -\frac{4\sigma}{3\kappa\rho} \nabla T^4. \quad (28)$$

Evaluated at the front, this gives

$$\rho e \dot{x}_F(y, t) = -\frac{4\sigma}{3\kappa\rho} \partial_x T^4 \big|_{x=x_F}. \quad (29)$$

This equation is the fundamental evolution law for the radiation front.

3.1 Mode expansion of the front

We expand the front position in transverse eigenmodes:

$$x_F(y, t) = \sum_{n=0}^{\infty} c_n(t) \cos(k_n y), \quad (30)$$

The time derivative is therefore

$$\dot{x}_F(y, t) = \sum_{n=0}^{\infty} \dot{c}_n(t) \cos(k_n y). \quad (31)$$

3.2 Evaluation of the radiative flux

From the reconstructed temperature field,

$$T^4(x, y) = \sum_n \cos(k_n y) (A_n e^{k_n x} + B_n e^{-k_n x}),$$

the longitudinal gradient is

$$\partial_x T^4 = \sum_n k_n \cos(k_n y) (A_n e^{k_n x} - B_n e^{-k_n x}). \quad (32)$$

Evaluating at $x = x_F$ and using the weak-curvature approximation

$$x_F(y, t) \approx c_0(t) + \mathcal{O}(\varepsilon),$$

we obtain to leading order

$$\partial_x T^4|_{x_F} \approx \sum_n k_n \cos(k_n y) (A_n e^{k_n c_0} - B_n e^{-k_n c_0}). \quad (33)$$

3.3 Evolution equations for the front modes

Substituting into the front balance equation (29) and projecting onto each transverse mode using orthogonality, we obtain a system of evolution equations:

Zeroth mode ($n = 0$):

$$\dot{c}_0(t) = \frac{D_M}{2} \left(1 + \frac{\varepsilon}{3}\right) \frac{\sqrt{\varepsilon}}{L \sinh\left(\frac{\sqrt{\varepsilon}}{L} c_0\right)} + \mathcal{O}(\varepsilon^2). \quad (34)$$

Higher modes ($n \geq 1$):

$$\dot{c}_n(t) = \frac{D_M}{2} \frac{4\varepsilon}{\pi^2} \frac{(-1)^n}{n^2} \frac{k_n}{\sinh(k_n c_0)} + \mathcal{O}(\varepsilon^2). \quad (35)$$

Defining the classical Marshak diffusion constant

$$D_M \equiv \frac{8\sigma T_s^4}{3\kappa\rho^2 e}, \quad (36)$$

3.4 Solution for the dominant front motion

Equation (34) can be integrated exactly, yielding

$$c_0(t) = \frac{L}{\sqrt{\varepsilon}} \cosh^{-1}\left(\frac{D\varepsilon t}{2L^2} + 1\right), \quad (37)$$

with the *modified diffusion constant*

$$D = D_M \left(1 + \frac{\varepsilon}{3}\right). \quad (38)$$

Expanding for small ε ,

$$c_0(t) \simeq \sqrt{Dt} - \frac{L}{12} \sqrt{\frac{2}{\varepsilon}} \left(\frac{D\varepsilon t}{2L^2} \right)^{3/2} + \dots, \quad (39)$$

showing explicitly the deviation from classical Marshak scaling.

Full front shape

To leading order, the front position is

$$x_F(y, t) = \frac{L}{\sqrt{\varepsilon}} \cosh^{-1} \left(\frac{D\varepsilon t}{2L^2} + 1 \right) \cos \left(\frac{\sqrt{\varepsilon}}{L} y \right) + \text{HOT}, \quad (40)$$

where “HOT” denotes higher-order transverse corrections.

Curvature radius

The curvature at the centerline $y = 0$ is given by

$$R_c^{-1} \approx - \left. \frac{\partial^2 x_F}{\partial y^2} \right|_{y=0}, \quad (41)$$

which yields

$$R_c(t) = \frac{L}{\sqrt{\varepsilon}} \frac{1}{\cosh^{-1} \left(\frac{D\varepsilon t}{2L^2} + 1 \right)}. \quad (42)$$

Thus, curvature increases monotonically with time.

Physical interpretation

- Energy is diverted sideways into the boundaries, reducing forward transport.
- The Marshak wave is no longer purely diffusive.
- The front is simultaneously:
 - **slowed** (longitudinal drag),
 - **bent** (transverse energy leakage),
 - **anisotropic** (geometry-dependent diffusion).
- In the limit $\varepsilon \rightarrow 0$:

$$x_F(t) \rightarrow \sqrt{D_M t},$$

recovering the classical Marshak solution.

Regime structure

notice that there is a problem if $R_c > L$, since then the front would curve back onto itself. This happens when

$$\frac{L}{\sqrt{\varepsilon}} \frac{1}{\cosh^{-1}\left(\frac{D\varepsilon t}{2L^2} + 1\right)} = L,$$

or equivalently we need

$$t < \frac{2L^2}{D\varepsilon} \left[\cosh\left(\frac{1}{\sqrt{\varepsilon}}\right) - 1 \right].$$

but since $\varepsilon \ll 1$, the RHS is very large, so this condition is usually satisfied in practice. Define the diffusion timescale

$$t_D \sim \frac{L^2}{D}. \quad (43)$$

Then:

- $t \ll t_D$: nearly planar, classical Marshak regime.
- $t \sim t_D$: onset of curvature and drag.
- $t \gg t_D$: strong bending, wall-dominated transport.

Final conceptual statement

A bounded Marshak wave is not a pure diffusion process. It is a geometrically constrained radiative transport phenomenon, in which lateral energy leakage produces an effective drag and curvature of the radiation front. The front dynamics are controlled by a single dimensionless parameter ε , which quantifies the competition between forward diffusion and sideways energy loss.

4 Conclusion

- Lossy walls bend and slow Marshak waves.
- 2D radiation diffusion fundamentally differs from 1D theory.
- Dynamics controlled by three parameters:
 - Tube half-width L
 - Wall-loss parameter ε
 - Modified diffusion constant D
- Theory explains experiments and simulations.

- Extensions proposed:
 - Cylindrical geometry
 - Temperature-dependent opacity $\kappa(T)$
 - Temperature-dependent internal energy $e(T)$
 - Coupled radiation-hydrodynamics